A Geometric and Bayesian Formulation of Autoregressive Language Models

Generative AI Perspectives

June 8, 2026

Abstract

This paper formalizes the mechanics of autoregressive Large Language Models (LLMs) by reframing their forward pass and pre-training objectives through the lens of Bayesian inference, high-dimensional geometry, and statistical mechanics. We show that the conditional probability distribution generated by the Transformer architecture can be explicitly decomposed into an intrinsic token prior and a contextual likelihood. We map these probabilistic components directly onto the network's structural blocks—specifically Multi-Head Attention (MHA) and Feed-Forward Networks (FFN)—and demonstrate how the final output token distribution is modulated by the statistical variance of the latent residual stream, acting as a dynamic inverse temperature. Finally, we formalize pre-training as a self-organizing geometric optimization that resolves these components into a stable semantic coordinate system.

1 Introduction and the Bayesian Framing

An autoregressive large language model operates over a finite vocabulary V. Given a context sequence of n tokens, $C = (t_1, t_2, \dots, t_n)$, where $t_i \in V$, the model defines a discrete conditional probability distribution over the vocabulary for the next token $w \in V$:

$$P(w \mid C) \tag{1}$$

While modern deep learning architectures evaluate this distribution directly using discriminative causal language modeling, we can invert this formulation using Bayes' theorem to expose the underlying generative dynamics:

$$P(w \mid C) = \frac{P(C \mid w) \cdot P(w)}{P(C)} \tag{2}$$

Because the denominator $P(C) = \sum_{v \in V} P(C \mid v)P(v)$ represents the marginal likelihood of the context sequence, it acts as a constant normalization factor across all candidate tokens at step $n+1$. Thus, the relative probability mass assigned to any token w is determined strictly by the numerator:

$$P(w \mid C) \propto P(C \mid w) \cdot P(w) \tag{3}$$

Where:

- $P(w)$ is the **unconditional prior probability** of the token w occurring in the language, independent of local context.
- $P(C \mid w)$ is the **contextual likelihood**, measuring how strongly the historical sequence C retroactively aligns with or necessitates the trailing token w.

2 Mathematical Decomposition of Likelihood $P(C \mid w)$

In a Transformer architecture, the context sequence C is transformed into a sequence of continuous vectors. Let $X \in \mathbb{R}^{n \times d}$ represent the input embedding matrix for the context, where d is the model dimension. The final hidden state at the terminal position n after L layers is denoted as $h_n^{(L)} \in \mathbb{R}^d$.

The unnormalized log-probabilities (logits) z_w for each token w are computed by projecting $h_n^{(L)}$ onto an unembedding matrix $U \in \mathbb{R}^{|V| \times d}$, such that $z_w = h_n^{(L)} \cdot u_w$, where $u_w \in \mathbb{R}^d$ is the target token's unembedding vector.

The log-likelihood $\log P(C \mid w)$ is shaped by the interaction of two foundational architectural components within each layer l: Multi-Head Attention and the position-wise Feed-Forward Network.

2.1 Multi-Head Attention as Contextual Routing

The Multi-Head Attention block at layer l and position n computes a dynamically weighted combination of all preceding token states. For a single attention head, let $W_Q^{(l)}, W_K^{(l)}, W_V^{(l)} \in \mathbb{R}^{d \times d_k}$ be the Query, Key, and Value projection matrices, respectively.

The terminal query vector is defined as $q_n^{(l)} = h_n^{(l-1)} W_Q^{(l)}$, while the keys and values for all historical positions $i \in \{1, \ldots, n\}$ are $k_i^{(l)} = h_i^{(l-1)} W_K^{(l)}$ and $v_i^{(l)} = h_i^{(l-1)} W_V^{(l)}$. The dynamic attention weights $\alpha_{n,i}^{(l)}$ formalize the conditional token-to-token dependencies:

$$\alpha_{n,i}^{(l)} = \frac{\exp\left(\frac{q_n^{(l)} \cdot k_i^{(l)}}{\sqrt{d_k}}\right)}{\sum_{j=1}^{n} \exp\left(\frac{q_n^{(l)} \cdot k_j^{(l)}}{\sqrt{d_k}}\right)} \tag{4}$$

The attention output $a_n^{(l)}$ is projected back into the residual stream via the output matrix $W_O^{(l)} \in \mathbb{R}^{d \times d}$. Its mathematical contribution to the log-likelihood of a target token w decomposes into a weighted sum of bilinear forms:

$$\log P(C \mid w)_{\text{MHA}} \sim \sum_{i=1}^{n} \alpha_{n,i}^{(l)} \left(h_i^{(l-1)} W_V^{(l)} W_O^{(l)} u_w \right) \tag{5}$$

This isolates two distinct mechanisms: $\alpha_{n,i}^{(l)}$ handles the contextual routing (determining which past tokens matter), while the dot product acts as a semantic alignment function between the historical features of token i and the target token w.

2.2 Feed-Forward Networks as Concept Memory Retrieval

Following the attention block, the updated hidden state $x_n^{(l)} = h_n^{(l-1)} + a_n^{(l)}$ enters the position-wise Feed-Forward Network (FFN). The FFN operates as a key-value associative memory. Let $W_1^{(l)} \in \mathbb{R}^{d \times d_{ff}}$ represent a bank of intermediate keys $\{k_m^{(l)}\}_{m=1}^{d_{ff}}$, and $W_2^{(l)} \in \mathbb{R}^{d_{ff} \times d}$ represent a bank of intermediate values $\{v_m^{(l)}\}_{m=1}^{d_{ff}}$.

Applying a non-linear activation function σ (e.g., GELU), the operation at the terminal position is:

$$\text{FFN}(x_n^{(l)}) = \sum_{m=1}^{d_{ff}} \sigma\left(x_n^{(l)} \cdot k_m^{(l)} + b_{1,m}^{(l)} \right) v_m^{(l)} \tag{6}$$

Projecting this operation onto the target token embedding u_w reveals the FFN's mathematical formulation of the likelihood:

$$\log P(C \mid w)_{\mathrm{FFN}} \sim \sum_{m=1}^{d_{ff}} \sigma\left(x_n^{(l)} \cdot k_m^{(l)} + b_{1,m}^{(l)}\right) \cdot (v_m^{(l)} \cdot u_w) \tag{7}$$

Here, $\sigma\left(x_n^{(l)} \cdot k_m^{(l)} + b_{1,m}^{(l)}\right)$ scales how strongly the synthesized context satisfies the m-th latent conceptual rule, while $(v_m^{(l)} \cdot u_w)$ dictates the historical co-occurrence correlation between that rule and the literal vocabulary token w.

3 Geometric Formulation of the Prior $P(w)$ and Logit Mechanics

The exact computation of the unconditional prior $P(w)$ requires marginalizing over the infinite space of all possible context sequences $\mathcal{C}$:

$$P(w) = \sum_{C \in \mathcal{C}} P(w \mid C) P(C) \tag{8}$$

Transformers approximate this expectation statically within their parameters via two geometric mechanics:

3.1 Unembedding Bias (b_U)

In architectures featuring an explicit unembedding bias $b_U \in \mathbb{R}^{|V|}$, the logit calculation is defined as $z_w = (h_n^{(L)} \cdot u_w) + b_{U,w}$. Under an uninformative context where $h_n \to \mathbf{0}$, the logit collapses directly to the scalar bias:

$$P(w) \approx \frac{\exp(b_{U,w})}{\sum_{v \in V} \exp(b_{U,v})} \tag{9}$$

3.2 Embedding Norms ($\|u_w\|$)

In bias-free architectures, the prior is encoded in the Euclidean magnitude of the token vector, $\|u_w\|$. Decomposing the final dot product geometrically yields:

$$z_w = \|h_n^{(L)}\| \|u_w\| \cos(\theta_w) \tag{10}$$

Where θ_w is the high-dimensional angle between the final context state and the token vector. Tokens with high baseline frequencies in the training corpus develop longer vector magnitudes $\|u_w\|$ during optimization, amplifying their baseline logit regardless of the contextual direction $\cos(\theta_w)$.

4 Statistical Mechanics of the Latent Stream

The remaining term in the geometric logit decomposition is the norm of the final hidden state, $\|h_n^{(L)}\|$. This component functions as a dynamic, context-dependent entropy controller.

4.1 Activation Variance as Inverse Temperature

Let $h_n = [x_1, x_2, \ldots, x_d]^T$. If we treat the individual channel activations as samples from an internal distribution with a mean driven to zero by layer normalization ($\mu_x \approx 0$), the cross-channel variance is:

$$\sigma^2 = \frac{1}{d}\sum_{i=1}^{d}(x_i - \mu_x)^2 \approx \frac{1}{d}\sum_{i=1}^{d} x_i^2 \tag{11}$$

Because the squared Euclidean norm is $\|h_n\|^2 = \sum_{i=1}^{d} x_i^2$, the vector magnitude can be expressed as a function of activation variance:

$$\|h_n\| = \sqrt{d \cdot \sigma^2} \tag{12}$$

When substituting this back into the vocabulary-wide Softmax function, $\|h_n\|$ maps perfectly to the **inverse temperature** parameter $\beta = \frac{1}{T}$ from statistical mechanics:

$$P(w \mid C) = \frac{\exp\big(\beta \cdot \|u_w\| \cos(\theta_w)\big)}{\sum_{v\in V} \exp\big(\beta \cdot \|u_v\| \cos(\theta_v)\big)}, \quad \text{where } \beta = \|h_n\| \tag{13}$$

- **High Contextual Certainty:** When the context contains a highly predictable pattern, successive attention heads and FFN layers write aligned information into the residual stream. This constructive interference inflates the cross-channel variance σ^2, causing $\|h_n\|$ to expand. This yields a high β, sharpening the distribution and lowering its entropy.
- **High Contextual Ambiguity:** When the context is entropic or transitions between distinct topics, architectural sub-layers write orthogonal or destructive vectors into the stream. The variance σ^2 collapses, shrinking $\|h_n\|$. This yields a low β, flattening the distribution into a high-entropy state.

4.2 Cosine Similarity as a Correlation Coefficient

Similarly, treating the dimensions of h_n and u_w as zero-mean random variables allows us to map the likelihood component $\cos(\theta_w)$ directly to the **Pearson correlation coefficient (r)** in feature space:

$$\cos(\theta_w) = \frac{\mathrm{Cov}(h_n, u_w)}{\sigma_{h_n}\sigma_{u_w}} = r_{h_n,u_w} \tag{14}$$

Thus, the unnormalized logit is exactly proportional to the total feature co-variance scaled by the model dimension: $z_w = \mathrm{Cov}(h_n, u_w) \cdot d$.

5 Statistical Description of Pre-Training

During inference, u_w is static. During pre-training, both the context state h_n and the vocabulary matrix U are discovered simultaneously via the minimization of the cross-entropy loss for a target next-token w^*:

$$\mathcal{L} = -\log \frac{\exp(h_n \cdot u_{w^*})}{\sum_{v\in V} \exp(h_n \cdot u_v)} \tag{15}$$

This framework can be understood as a dual-force geometric optimization process:

5.1 The Mutual Attraction Force

The gradient with respect to the correct token vector u_{w^*} acts as an attractive force:

$$\frac{\partial \mathcal{L}}{\partial u_{w^*}} = -\big(1 - P(w^* \mid C)\big)h_n \tag{16}$$

This force dynamically rotates u_{w^*} and the context representation h_n toward one another, maximizing their directional correlation $\cos(\theta_{w^*})$. Simultaneously, if a token w^* is consistently validated across diverse contexts, its vector norm $\|u_{w^*}\|$ is driven upward to minimize loss across the corpus, mapping out its global prior $P(w)$.

5.2 The Global Repulsion Force

Conversely, the denominator of the Softmax generates a contrastive repulsion force against all incorrect tokens $v \neq w^*$:

$$\frac{\partial \mathcal{L}}{\partial u_v} = P(v \mid C) h_n \tag{17}$$

This pushes the context state h_n away from incorrect vocabulary vectors, forcing them toward orthogonality ($\cos(\theta_v) \to 0$).

5.3 Emergence of Distributional Semantics

Because the network minimizes this loss across billions of tokens, semantic clustering emerges naturally. If two distinct tokens w_1 and w_2 share high contextual substitutionality, they will repeatedly receive identical directional updates from shared context states h_n. This self-organizing system settles into a geometric steady-state where vocabulary vectors discover a coordinate system whose spatial proximity mirrors latent semantic relationships:

$$\cos(\theta_{w_1,w_2}) \to 1 \iff \mathcal{C}_{w_1} \approx \mathcal{C}_{w_2} \tag{18}$$

6 Conclusion

By formalizing the Large Language Model as a Bayesian engine, we transition from a qualitative understanding of neural network behaviors to an exact geometric framework. Multi-Head Attention and Feed-Forward Networks act in tandem to iteratively construct a context vector whose final magnitude represents the model's statistical certainty (β), and whose orientation represents structural feature configurations. The final token prediction is thus a direct measure of the Pearson correlation between this dynamic contextual variance and the self-organized coordinate landscape of the vocabulary embeddings.